Wenfeng Wang

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Education

University of Pennsylvania Philadelphia, PA
M.S.E. in Electrical Engineering GPA: 3.72/4.00 *Aug 2025 – May 2027 (expected)*
- Coursework: Machine Learning, Graph Neural Networks, Statistics for Data Science, Digital Integrated Circuits & VLSI, Artificial Intelligence*, Linear Systems Theory* (*in progress)

Jilin University Changchun, China
B.S. in Physics GPA: 3.80/4.00 *Sep 2020 – Jul 2024*
- Coursework: Electrodynamics, Quantum Mechanics, Thermal Physics, Mathematical Analysis, Linear Algebra, Probability

Publications

- **Wenfeng Wang**, Jianrui Shi, Mingang Guo, Jiayuan Ma, Yuqi Lu, Jiajun Sun, Guojiang Zhao, Jing Li. Trace-to-Architecture: Capability Bottleneck Diagnosis for Harness Revision. *arXiv preprint* arXiv:2610.16725, 2026. Under review at ACL Rolling Review.

Research Experience

Haz Lab, Drexel University Philadelphia, PA
Research Volunteer, with Prof. Li (Harry) Zhang *Oct 2026 – present*
- Studying when LLM agents should compile recurring requests into reusable programs (formalization); surveyed about 100 related works on skill libraries, program caching, and agent memory.

Trace-to-Architecture: Capability Bottleneck Diagnosis for Harness Revision 2026 – present
First author
- Proposed T2A, a framework that uses execution trajectories to decide what to change in an LLM agent's harness when it fails, and when those changes should apply.
- Built tooling that groups trajectories by observable features, letting a development agent diagnose failures with evidence from both successful and failed tasks.
- Designed validation that checks not only whether target tasks improve, but also whether selected successful tasks stay successful and whether the edits are actually triggered.
- Across ALFWorld, ScienceWorld, and AppWorld, achieved the highest aggregate score for all three task agents, exceeding the strongest baselines by 6.2–30.3 points; integrating T2A's diagnosis and validation into Meta-Harness raised its mean ScienceWorld score by 21.6 points.

Listening or Writing? Speech Recognition for Clinical Notes 2026
Independent project; implemented with Claude Code *github.com/Fw93810/medical-asr-eval*
- Evaluated six open-weight ASR models and three local LLM note writers on PriMock57 mock consultations with a blinded fact-level judge; in quiet audio the note writer moved fact recall by 8–12 points versus 1–3 for ASR errors, while 0 dB babble made the ASR's cost comparable.

University of Chinese Academy of Sciences Beijing, China
Research Intern *Jul 2024 – Aug 2025*
- Studied automatic evaluation of text-to-image generation (CLIPScore, VQAScore, BLIP-2), focusing on how well metrics capture spatial relations between objects.
- Explored a 3D-aware evaluation dimension: recovered scene structure with monocular 3D reconstruction and fed text, image, and 3D tokens jointly into a language model to assess spatial orientation and agent–patient interactions.
- Reproduced and deployed vision and generation models (VGG16/19, YOLOv5, Omost) in PyTorch.

Introduction to VLSI Research Program Remote
Supervised by Prof. Dejan Marković, UCLA *Jul 2023 – Aug 2023*

Nanoelectronics, Optoelectronics and AI Research Seminar Remote
Supervised by Prof. Deep Jariwala, University of Pennsylvania *Feb 2023 – Apr 2023*

Awards

Jilin University Scholarships, Second Prize (2022) and Third Prize (2020); Yan Jici Scholarship, Institute of Physics, Chinese Academy of Sciences (2021)

Skills

Programming: Python, PyTorch, C, LaTeX **Languages:** Chinese (native), English (fluent)